DC Electronic Load Based on 485 Bus(Score Table) V6.3

SN: _____ Name: _____ StuNo: _____ Class: _____

<table>
<tr><th>Function</th><th>Test Items and Results Check Count#:____</th><th>Score</th></tr>
<tr><td rowspan="6">Basic Part (50 Points)</td><td>1) HMI: (10') StuInfo Display(PCB & Software): ☐Yes ☐No[Stop Check]
☐Keys Function(3') ☐Multi Level Menu(3') ☐LED Indicator(1')
☐Buzzer(1') ☐Parameter Storage on Power-down(2')</td><td></td></tr>
<tr><td>2) CC Mode: Current Range 100mA～3000mA，resolution 10mA，Accuracy ≤ ±1%(20') Actual Resolution:_____mA
<table>
<tr><td>Set Current (A)</td><td>0.100</td><td>1.000</td><td>3.000</td></tr>
<tr><td>Measured Current (A)</td><td></td><td></td><td></td></tr>
<tr><td>Accuracy (%)</td><td></td><td></td><td></td></tr>
</table></td><td></td></tr>
<tr><td>3) CC Mode: When the voltage at both ends of the load is from 5V to 3V, the absolute value change of output current is required to be less than 1%（Set Current 1.000A）(5')
<table>
<tr><td>Port Voltage (V)</td><td>5.000</td><td>4.000</td><td>3.000</td></tr>
<tr><td>Measured Current (A)</td><td></td><td></td><td></td></tr>
<tr><td>Accuracy (%)</td><td></td><td></td><td></td></tr>
</table></td><td></td></tr>
<tr><td>4) Open circuit set function(Iout=0)(5') Iout(Min)=_____mA ☐Yes ☐No</td><td></td></tr>
<tr><td>5) Over voltage protection(≥5.5V±0.2V) (5'): ☐Yes ☐No</td><td></td></tr>
<tr><td>6) Over temperature protection(Iout≥1A,Fan on) (5'): ☐Yes ☐No</td><td></td></tr>
<tr><td rowspan="4">Advanced Part (30 Points)</td><td>1) Real-time voltage measure and display(input port), Accuracy≤±1%(8'):
Displayed voltage_____V, Measured voltage_____V, Error_____%</td><td></td></tr>
<tr><td>2) Real-time current measure and display, Accuracy≤±1%(8'):
Displayed current_____A, Measured current_____A, Error_____%</td><td></td></tr>
<tr><td>3) Automatic measure of Voltage Load Regulation Rate，Error≤±1%(6'):
Measured voltage (Iout=0A)_____V, Measured voltage (Iout=1A)_____V
VLRR Measured_____%, Displayed value_____%, Error_____%</td><td></td></tr>
<tr><td>4) ModBus-RTU 485 Bus(Read and Write Parameters) (8'): bps: 9600
☐Node Addr(2digits-StuNo) ☐Read ☐Write ☐SCADA(Supervisory Control and Data Acquisition，监控与数据采集，组态软件)</td><td></td></tr>
<tr><td rowspan="3">Extended Part (10 Points)</td><td>1) Constant Resistance Mode (3'): set____Ω：_____Ω ☐Yes ☐No</td><td></td></tr>
<tr><td>2) Constant Power Mode (3'): set____W：_____W ☐Yes ☐No</td><td></td></tr>
<tr><td>3) Others (4'): ☐Others_____
☐System Fault Indication ☐MOSFET Over-temperature Protection
☐Color-screen Display ☐Self-designed PC Software ☐MCU_____</td><td></td></tr>
<tr><td rowspan="2">Conception (10 Points)</td><td>1) Design standardization (5')
PCB layout, wiring, silk screening, flying leads, solder joints, pin trimming, etc</td><td></td></tr>
<tr><td>2) Consideration of project sustainability factors (5')
Material recycling</td><td></td></tr>
<tr><td colspan="2">Check Date:_____Y__M__D__H Teacher:_____ Total（100 Points）</td><td></td></tr>
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